

PC-Weather: Preserve-and-Compose Weather Editing via Intrinsic Anchoring and Region-Aware Velocity Composition

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Abstract—Weather editing for traffic scenes seeks to synthesize target adverse conditions while preserving the geometry, object identities, and spatial layout of the original scene. However, existing methods typically preserve content by directly conditioning on the source image. This representation conflates material appearance with source-weather illumination, making it difficult to remove lighting effects tied to the original condition. This limitation is further exacerbated when editing is guided by a simple instruction, which cannot specify the distinct weather responses required across different regions. To address these coupled limitations, we present PC-Weather, a training-free framework that combines intrinsic anchoring for global preservation with region-aware velocity composition for localized weather adaptation. For preservation, *Intrinsic Anchoring* estimates an albedo map from the source image and encodes it as a distinct reference-token sequence alongside the source-image tokens in the global velocity stream. For regional adaptation, *Region-Aware Velocity Composition* refines the global weather instruction into region-specific conditions and evaluates their velocity fields at the same latent state as the global field. The global field defines a common transport direction, and each regional field can alter it only through a residual correction within its semantic mask. The magnitude of this masked correction is then regulated by a token-wise discrepancy gate. On traffic-scene images from STREET-6K, BDD100K, and Cityscapes, PC-Weather achieves the highest instruction-success score of 9.640. Across scratch-training and fine-tuning protocols, PC-Weather-synthesized data yield the highest average mAP_{50:95} and mAP₅₀ in cross-dataset vehicle detection on the DAWN and ACDC test sets.

Index Terms—Weather Transfer, Controllable Image Editing, Flow Matching.

I. INTRODUCTION

WEATHER editing for traffic scenes seeks to synthesize target conditions such as rain, snow, and fog while preserving task-relevant geometry, object identities, and spatial layout. This capability provides a scalable means of augmenting the rare adverse-weather data needed to train and evaluate robust autonomous-driving perception systems. However, synthesized images are useful for this purpose only when they introduce the requested environmental changes without invalidating the scene content or its associated labels. Although diffusion-based image editing [1]–[7] has achieved impressive visual quality, balancing effective weather transformation with reliable content preservation remains challenging.

Recent weather-editing methods have improved realism, diversity, and controllability through instruction-guided generation, spatial and semantic guidance, video diffusion, and

reconstruction-based simulation [8]–[11]. Nevertheless, these complementary controls do not jointly provide an illumination-reduced material reference and explicit category-dependent weather responses for arbitrary single-image editing. As illustrated in Fig. 1, this gap produces two characteristic failures: preservation that is too strong leaves weather-dependent content semantically inconsistent with the target condition, whereas aggressive global rewriting can distort object shapes, scene boundaries, or material identities.

Our key insight is that weather editing is not a uniform style-transfer problem, but a spatially varying balance between preservation and reconstruction. The required balance differs across categories and may even coexist within the same region: road geometry and lane boundaries should remain fixed, while road reflectance or snow accumulation may need to change; similarly, vehicle identity and shape should be preserved while illumination, reflections, or visibility adapt to the target weather. This perspective exposes two coupled ambiguities in existing editing pipelines. First, the source RGB image entangles material reflectance with weather-dependent illumination, so source-only preservation guidance may either retain obsolete lighting cues or allow material colors to drift. Second, a global instruction such as “make the scene snowy” specifies the overall condition but under-describes how individual semantic categories should respond.

To address these ambiguities, we propose **PC-Weather**, a training-free framework that separates global preservation from region-specific weather editing in the velocity space of a pretrained rectified-flow model. First, *intrinsic anchoring* estimates an albedo map using a pretrained intrinsic decomposition model [14] and encodes it as a separate reference alongside the source image. Rather than treating the estimate as an exact physical decomposition, we use it as a comparatively illumination-reduced cue to material identity and structural boundaries for the global preservation stream [15]–[18]. Second, an off-the-shelf large language model refines the global weather request into category-specific instructions, whose effects are restricted by corresponding semantic masks. At each shared latent state, the albedo-anchored global stream defines a reference velocity, and each regional stream contributes only the residual induced by its category-specific condition. Because all fields are evaluated at the same state with the same frozen backbone, these residuals directly characterize condition-induced changes in the instantaneous transport direction. Our *Region-Aware Velocity Composition* applies them only within their semantic masks and modulates their magnitude with a token-wise discrepancy gate. Thus, regional fields are not separately integrated into local image trajectories; they can only deform a single globally anchored update.

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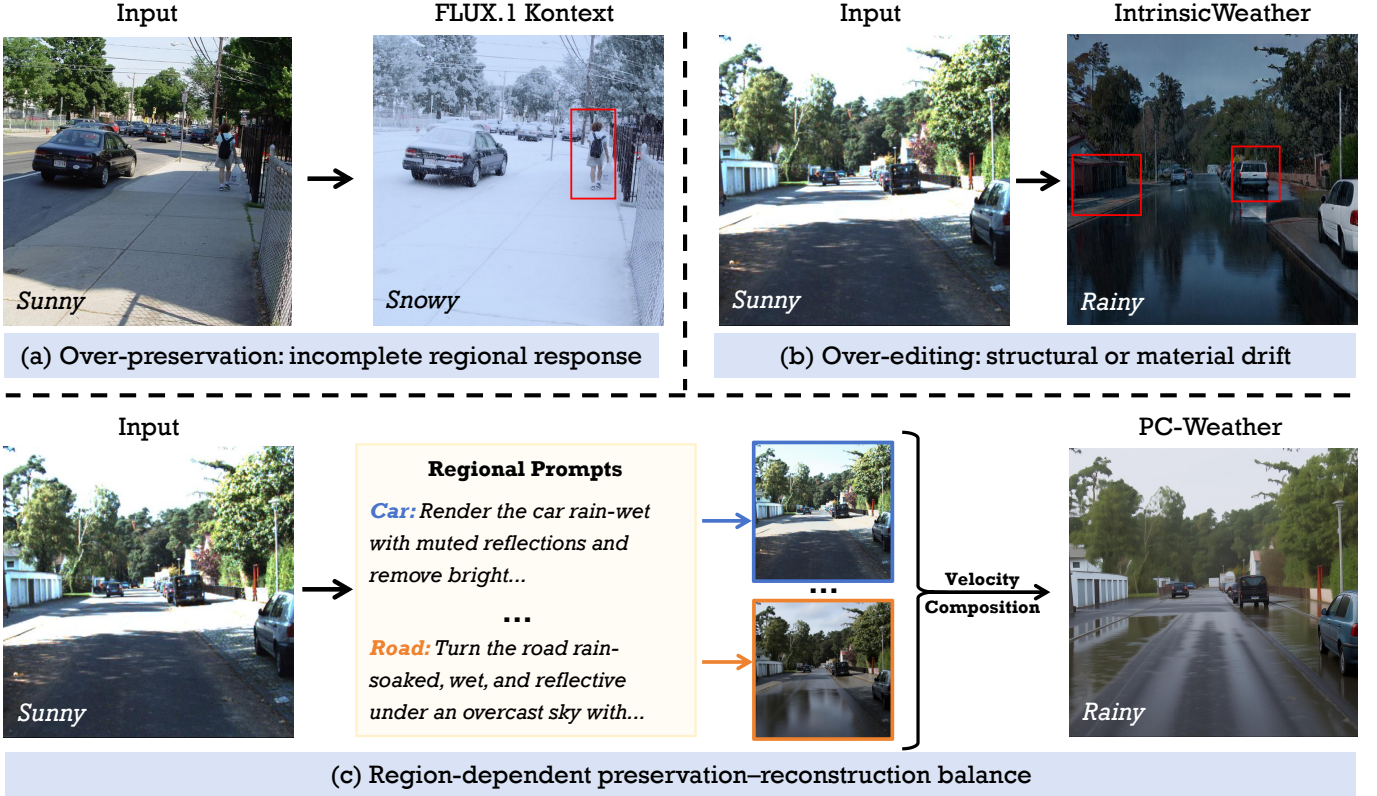


Fig. 1. **Motivation for region-dependent weather editing.** (a) FLUX.1 Kontext [12] over-preserves the source content, retaining the pedestrian’s clear-weather appearance in the snowy scene. (b) IntrinsicWeather [13] aggressively reconstructs the scene, causing visible drift in building appearance and vehicle identity (red boxes). (c) Given the same source image as in (b), PC-Weather composes region-specific velocity corrections with the globally anchored velocity, producing a reflective rainy road while better preserving vehicle identities and scene layout. The complete framework is shown in Fig. 2.

This construction maintains a coherent scene-level trajectory while permitting stronger, region-appropriate changes where needed, without model training or per-image optimization.

We evaluate PC-Weather on 492 real-world traffic-scene images sampled from STREET-6K, BDD100K, and Cityscapes using complementary automated metrics, a forced-choice user study, and cross-dataset vehicle detection on DAWN and ACDC. Across these evaluations, PC-Weather consistently balances target-weather fidelity with source-scene preservation and achieves a 53.9% mean preference rate in the user study. The downstream results further show that the synthesized data transfers effectively to real adverse-weather perception across detector generations and training protocols.

In summary, our contributions are as follows:

- We formulate traffic-scene weather editing as a region-dependent preservation–reconstruction problem, motivating category-dependent adaptation that coordinates weather transformation with label-relevant content preservation.
- We introduce **PC-Weather**, a training-free framework that combines intrinsic anchoring with discrepancy-gated, region-aware velocity composition to coordinate global content preservation and category-specific weather responses.
- We provide comprehensive evaluations of perceptual quality, human preference, and downstream detection utility, demonstrating consistent benefits across multiple

traffic-scene and real adverse-weather benchmarks.

II. RELATED WORK

A. Weather Synthesis for Driving Scenes

Weather synthesis introduces a target weather condition while preserving the geometry, object identities, and layout of the source scene. Unlike adverse-weather restoration, it must jointly alter atmospheric visibility, illumination, surface appearance, and weather-dependent semantics. This requirement is especially important in traffic scenes, where changes to vehicles, lane markings, signs, or road boundaries can invalidate labels for downstream perception.

Recent methods have improved the realism and controllability of weather translation. InstructWT [8] supports unified multi-weather editing with controllable weather intensity, Style Nursing [10] introduces spatial and semantic guidance for traffic-scene transfer, and WeatherWeaver [9] learns controllable weather synthesis and removal for videos. These methods offer complementary prompt-, feature-, or temporal-level controls. However, global weather control and generic content guidance do not explicitly separate weather-varying illumination from material cues while assigning different target responses to semantic categories in a single edited image. PC-Weather instead addresses this setting by coordinating an illumination-reduced preservation reference and category-dependent responses within one inference-time trajectory.

B. Intrinsic and 3D Physically Grounded Weather Editing

Explicit representations provide another route to weather editing. RGB $\leftrightarrow$ X [19] and IntrinsicEdit [20] use material- and lighting-aware representations for image manipulation, while IntrinsicWeather [13] estimates intrinsic maps and performs weather-conditioned forward rendering. RGB $\leftrightarrow$ X focuses on intrinsic-channel decomposition and synthesis for indoor scenes, whereas IntrinsicEdit enables precise local editing and relighting in intrinsic space. These intrinsic-space methods make the roles of material, geometry, and illumination more explicit than pixel-space editing, but their inverse and forward components are learned for the representations and data regimes they target. At the scene level, ClimateNeRF [21] and WeatherEdit [11] use reconstructed 3D/4D representations for physically meaningful and view-consistent weather effects; their pipelines consequently require multi-view or multi-frame capture, camera information, and scene reconstruction. PC-Weather takes a lighter single-image approach: estimated albedo is a soft, illumination-reduced reference for global preservation rather than an exact decomposition, leaving the regional streams free to synthesize target-weather changes.

C. Training-Free and Flow-Based Image Editing

Prompt-to-Prompt controls cross-attention maps, Plug-and-Play Diffusion Features injects guidance features and attention, and MasaCtrl uses mutual self-attention to query source content during editing [2], [3], [5]. MultiDiffusion [22] fuses condition-specific diffusion predictions under spatial constraints. Flow Matching and Rectified Flow make the generation transport explicit [23], [24]. FlowEdit constructs an inversion-free ODE that directly transports a source distribution toward a target distribution, whereas FlowAlign regularizes the editing trajectory with a flow-matching objective to improve source consistency [25], [26]. These methods provide general inference-time controls, but do not formulate weather-specific, category-dependent deviations around a privileged preservation field. In contrast, PC-Weather designates an albedo-anchored velocity as a scene-level reference and permits category-conditioned fields to affect sampling only through semantic-mask-restricted residuals evaluated at the same latent state. The discrepancy gate therefore regulates semantically specified local deviations around one common trajectory, rather than independently generating separate regional trajectories.

III. METHOD

We present **PC-Weather**, a training-free framework that accounts for the spatially heterogeneous effects of weather in traffic scenes. Our key idea is to separate content preservation from region-specific weather editing in the velocity space of a pretrained rectified-flow model. As illustrated in Fig. 2, a global stream uses the source image and its albedo map to preserve scene layout, object geometry, and material appearance, while regional streams use category-specific instructions to model how different semantic regions should respond to the target weather. At each sampling step, we compare each

regional velocity with the global velocity and use their token-wise difference to adaptively control the regional update within the corresponding semantic mask. This design permits substantial weather changes where necessary while maintaining the integrity of safety-critical scene content, without requiring additional training.

A. Flow-Based Editing Formulation

Training-free weather editing requires changing how source content is transported toward the target condition without modifying the pretrained model. Rectified-flow models make this transport explicit through their velocity fields, providing a natural interface for inference-time control. We therefore build PC-Weather on FLUX.1 Kontext [12], a context-aware rectified-flow model conditioned on a source image and a textual editing instruction. Its generation trajectory follows an ordinary differential equation (ODE) in the latent token space:

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}_\theta(\mathbf{x}_t, t; \mathcal{C}) \quad (1)$$

where $\mathbf{x}_t \in \mathbb{R}^{L \times C}$ contains L target-image tokens with channel dimension C , $t \in [0, 1]$ is the continuous noise level, and $\mathcal{C}$ denotes the visual and textual conditions. Sampling starts from $\mathbf{x}_{t_0} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and follows a decreasing schedule $1 = t_0 > t_1 > \dots > t_N = 0$.

Using the standard explicit Euler discretization, each sampling step is given by

$$\mathbf{x}_{t_{i+1}} = \mathbf{x}_{t_i} + (t_{i+1} - t_i)\mathbf{v}_i \quad (2)$$

where $\mathbf{v}_i$ denotes the composed velocity defined in Sec. III-D. PC-Weather modifies only this instantaneous transport direction by constructing $\mathbf{v}_i$ from complementary condition-specific predictions; the model parameters, noise schedule, and native ODE solver remain unchanged. The proposed control is therefore fully compatible with the pretrained sampling process and requires no task-specific optimization or fine-tuning.

B. Intrinsic Anchoring

Weather editing should alter illumination and atmospheric appearance while preserving the underlying materials. However, the source RGB image entangles material reflectance with weather-dependent illumination. Using it alone as the preservation reference therefore creates an ambiguity: the edited image may retain the source-weather lighting, or its material colors may drift with the target weather. We reduce this ambiguity by introducing an estimated albedo map as an illumination-reduced reference for the global preservation stream.

Given a source image I , we use the pretrained Marigold-IID-Appearance model [14] and retain its albedo prediction:

$$A = \mathcal{G}_{\text{IID}}(I), \quad (3)$$

where $\mathcal{G}_{\text{IID}}$ denotes the frozen albedo estimator. We do not assume A to be an exact physical decomposition. Instead, it provides a comparatively illumination-invariant cue to the material colors and structural boundaries of the source scene,

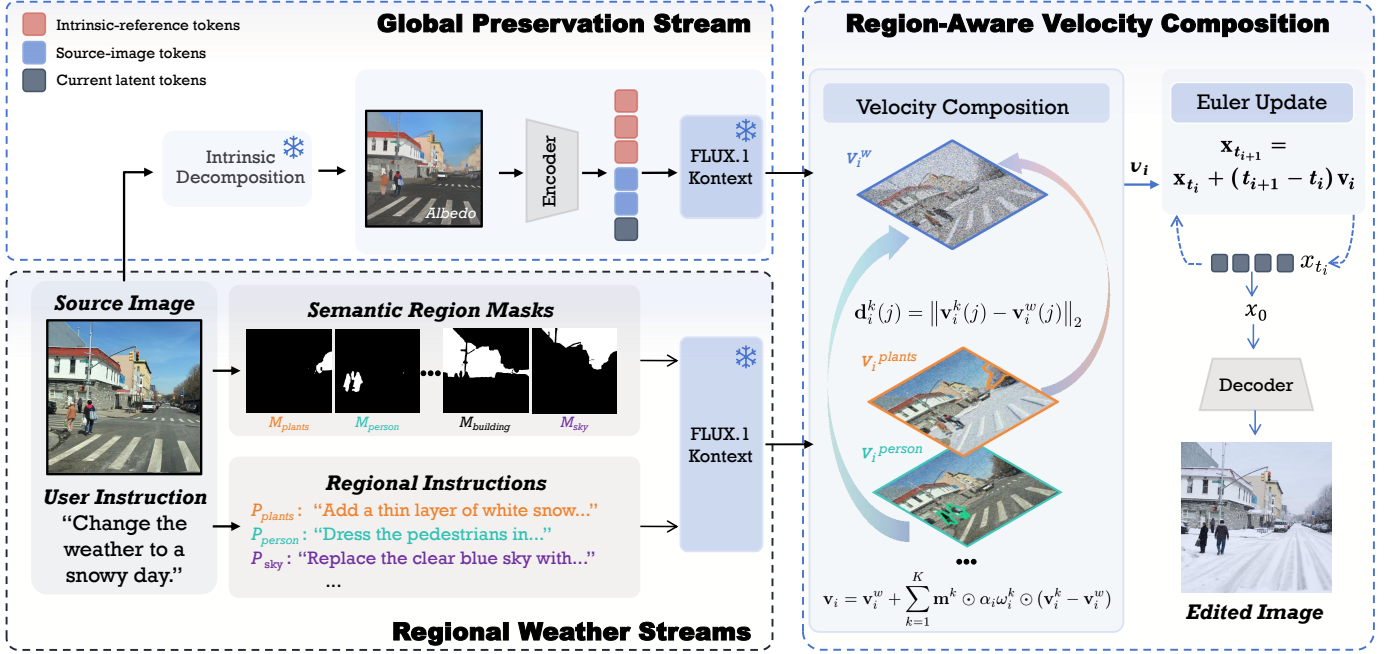


Fig. 2. **Overview of PC-Weather.** Given a source image I and a target-weather instruction p^w , PC-Weather constructs complementary global and regional velocity fields within a frozen rectified-flow backbone. An intrinsic decomposition model [14] estimates an albedo map A as an illumination-reduced reference for the global preservation stream. Meanwhile, an LLM refines p^w into regional instructions $\{p^k\}_{k=1}^K$, and semantic segmentation supplies their corresponding masks $\{M^k\}_{k=1}^K$. At each sampling step, the global stream predicts a reference velocity $\mathbf{v}_i^w$ from the source image, the target-weather instruction, and the intrinsic reference, whereas the regional streams predict $\{\mathbf{v}_i^k\}_{k=1}^K$ from the source image and regional instructions without the intrinsic reference. Region-Aware Velocity Composition restricts the regional-global velocity corrections to their semantic supports and weights them according to their token-wise discrepancies $d_i^k(j)$, yielding the composed velocity $\mathbf{v}_i$ for the Euler update.

serving as a preservation prior rather than a ground-truth physical constraint.

We represent the source observation and the albedo prior as separate contextual token sequences:

$$\mathbf{s} = \text{Enc}_{\text{seq}}(I), \quad \mathbf{r} = \text{Enc}_{\text{ref}}(A), \quad (4)$$

where $\mathbf{s}$ and $\mathbf{r}$ denote the source-image and intrinsic-reference tokens, respectively. Distinct positional identifiers preserve their separate roles when both sequences interact with the target tokens through the joint attention blocks of FLUX.1 Kontext. Consequently, the albedo reference supplies persistent cues about material identity and structural boundaries throughout sampling without prescribing the illumination of the edited image.

The intrinsic reference is applied only to the global preservation stream; each regional stream retains $\mathbf{s}$ but omits $\mathbf{r}$. This deliberate asymmetry prevents the preservation prior from suppressing category-dependent weather synthesis: the globally anchored trajectory provides a stable structural reference, while the regional streams remain free to model stronger semantic changes.

C. Regional Instruction Refinement

Weather changes are spatially heterogeneous. For example, snow accumulates on exposed surfaces, rain changes road reflectance, and fog reduces the contrast of distant regions. A single global instruction does not specify these category-dependent responses. We therefore query an off-the-shelf

LLM, Qwen3 [27], with the global instruction p^w and the semantic label c_k of each selected region. The returned regional instruction p^k concisely describes the desired changes in texture, color, illumination, or physical state while preserving the region’s object identity.

The refinement is designed to reflect two broad response tendencies rather than impose an additional region-classification stage. For **geometry-sensitive regions**, such as roads and vehicles, the refined instructions emphasize surface and illumination changes while preserving shape and category. For **weather-sensitive regions**, such as sky and vegetation, they allow stronger appearance or state changes. The global and regional instructions are encoded by the frozen T5 encoder as $\mathbf{e}^w = \text{T5}(p^w)$ and $\mathbf{e}^k = \text{T5}(p^k)$, respectively.

In parallel, a pretrained Mask2Former [28] predicts a semantic label map $S = \mathcal{G}_{\text{seg}}(I)$. For each selected category c_k , we construct a binary mask

$$M^k(u) = \mathbb{I}[S(u) = c_k], \quad (5)$$

$$u \in \{1, \dots, H\} \times \{1, \dots, W\},$$

where $S \in \{1, \dots, C_s\}^{H \times W}$, C_s is the number of segmentation categories, and $\mathbb{I}[\cdot]$ is the indicator function. We project each mask onto the latent-token grid and flatten it as $\mathbf{m}^k \in \{0, 1\}^{L \times 1}$. Together, $(\mathbf{e}^k, \mathbf{m}^k)$ specifies both the weather response assigned to category c_k and the spatial support over which it is allowed to act.

D. Region-Aware Velocity Composition

The global and regional conditions defined above favor complementary but potentially conflicting trajectories. The globally anchored field preserves scene structure and material identity, but may attenuate category-specific weather changes; conversely, the regional fields can express stronger local responses, but applying them indiscriminately may disrupt global consistency. Region-Aware Velocity Composition (RVC) reconciles these objectives at each shared latent state by treating the global velocity as a reference and the regional-global differences as spatially restricted corrections. For clarity, we omit the backbone’s shared guidance inputs from the following notation.

Global preservation stream: Conditioned on the global weather instruction and both the source and intrinsic references, the reference velocity is

$$\mathbf{v}_i^w = f_\theta(\mathbf{x}_{t_i}, t_i; \mathbf{e}^w, \mathbf{s}, \mathbf{r})|_{1:L} \quad (6)$$

where f_θ is the frozen FLUX.1 Kontext transformer and $|_{1:L}$ retains only the target-image velocity tokens. This field provides the baseline transport for maintaining scene-level consistency.

Regional weather streams: For region k , the corresponding weather field is conditioned on the regional instruction and source-image context, while excluding the intrinsic reference:

$$\mathbf{v}_i^k = f_\theta(\mathbf{x}_{t_i}, t_i; \mathbf{e}^k, \mathbf{s})|_{1:L}, \quad k = 1, \dots, K. \quad (7)$$

This asymmetry prevents the global preservation prior from over-constraining local weather reconstruction. Since $\mathbf{v}_i^w$ and $\mathbf{v}_i^k$ are evaluated at the same $\mathbf{x}_{t_i}$ and t_i with the same frozen backbone, their difference represents the change in transport direction induced by the regional condition. RVC does not separately integrate regional fields into independent image trajectories; instead, it parameterizes each local effect as a residual around the privileged global reference, $\Delta \mathbf{v}_i^k = \mathbf{v}_i^k - \mathbf{v}_i^w$.

Applying this regional correction uniformly throughout a mask assumes that all tokens require the same degree of change and can produce abrupt transitions. We instead quantify the correction magnitude at each token:

$$d_i^k(j) = \|\Delta \mathbf{v}_i^k(j)\|_2, \quad j = 1, \dots, L, \quad (8)$$

where j indexes a latent token. Following the general principle of reconciling concurrent generative paths [22], we map this discrepancy to a smooth, monotonically increasing gate

$$\omega_i^k(j) = \sigma(\beta d_i^k(j)) \quad (9)$$

where β controls the sensitivity to velocity disagreement. When the two fields agree, the correction itself remains small; as their disagreement increases, the gate gives the regional trajectory greater influence. With a step-dependent editing coefficient $\alpha_i \in [0, 1]$, the composed velocity is

$$\mathbf{v}_i = \mathbf{v}_i^w + \sum_{k=1}^K \mathbf{m}^k \odot \alpha_i \omega_i^k \odot \Delta \mathbf{v}_i^k \quad (10)$$

Algorithm 1 PC-Weather Sampling with RVC

Require: source image I , instructions $p^w, \{p^k\}_{k=1}^K$, semantic masks $\{M^k\}_{k=1}^K$, initial noise $\mathbf{x}_{t_0}$
Require: editing coefficients $\{\alpha_i\}_{i=0}^{N-1}$ with $\alpha_i \in [0, 1]$, sensitivity $\beta > 0$

- 1: $A \leftarrow \mathcal{G}_{\text{IID}}(I)$; $\mathbf{s} \leftarrow \text{Enc}_{\text{seq}}(I)$; $\mathbf{r} \leftarrow \text{Enc}_{\text{ref}}(A)$
- 2: $\mathbf{e}^w \leftarrow \text{T5}(p^w)$; $\mathbf{e}^k \leftarrow \text{T5}(p^k)$ for $k = 1, \dots, K$
- 3: $\mathbf{m}^k \leftarrow \text{Flatten}(\text{Downsample}_{\text{NN}}(M^k))$ for $k = 1, \dots, K$
- 4: **for** $i = 0$ **to** $N - 1$ **do**
- 5: $\mathbf{v}_i^w \leftarrow f_\theta(\mathbf{x}_{t_i}, t_i; \mathbf{e}^w, \mathbf{s}, \mathbf{r})|_{1:L}$
- 6: $\mathbf{v}_i \leftarrow \mathbf{v}_i^w$
- 7: **for** $k = 1$ **to** K **do**
- 8: $\mathbf{v}_i^k \leftarrow f_\theta(\mathbf{x}_{t_i}, t_i; \mathbf{e}^k, \mathbf{s})|_{1:L}$
- 9: $d_i^k(j) \leftarrow \|\mathbf{v}_i^k(j) - \mathbf{v}_i^w(j)\|_2$; $\omega_i^k(j) \leftarrow \sigma(\beta d_i^k(j))$ for $j = 1, \dots, L$
- 10: $\mathbf{v}_i \leftarrow \mathbf{v}_i + \mathbf{m}^k \odot \alpha_i \omega_i^k \odot (\mathbf{v}_i^k - \mathbf{v}_i^w)$
- 11: **end for**
- 12: $\mathbf{x}_{t_{i+1}} \leftarrow \mathbf{x}_{t_i} + (t_{i+1} - t_i) \mathbf{v}_i$
- 13: **end for**
- 14: **return** edited image $\hat{I} \leftarrow \text{Dec}(\mathbf{x}_{t_N})$

Here, $\mathbf{m}^k$ restricts each residual correction to its semantic support, α_i controls the overall regional editing strength at step i , and ω_i^k adapts that strength across tokens. The formulation has an explicit invariance: for every token outside the selected semantic supports, $\mathbf{v}_i(j) = \mathbf{v}_i^w(j)$ exactly. Within region k , the category-specific condition can only deform the globally anchored transport toward $\mathbf{v}_i^k$, with its extent controlled by $\alpha_i \omega_i^k$. This preserves a globally consistent base trajectory while allowing heterogeneous weather responses where they are required.

IV. EXPERIMENTS

A. Experimental Setup

a) **Datasets.**: We randomly sampled a total of 492 real-world traffic-scene images from three distinct datasets: (i) the STREET-6K benchmark from Style Nursing [10], (ii) BDD100K [29], and (iii) Cityscapes [30].

b) **Baselines.**: To evaluate the efficacy of PC-Weather, we conduct extensive comparisons with a representative suite of text-driven diffusion baselines: InstructPix2Pix [1], RGB $\leftrightarrow$ X [19], FLUX.1 Kontext [12], ICEdit [31], and IntrinsicWeather [13]. All these methods rely on text prompts as guidance.

c) **Evaluation Metrics.**: Given that weather and seasonal transitions involve significant alterations in lighting and atmospheric effects, traditional metrics such as PSNR, SSIM [32] and LPIPS [33] often misinterpret the legitimate cross-domain variations as reconstruction errors. Furthermore, they are limited in evaluating high-level semantic integrity and physical realism, which makes them insufficient for characterizing the performance of controllable generative frameworks in complex traffic environments.

To ensure a comprehensive and objective evaluation, we use an MLLM-based evaluation setup inspired by OmniEdit [34] and leverage a multimodal large language model (MLLM), Qwen3-VL-Plus [35], as an automated evaluator. We quantify

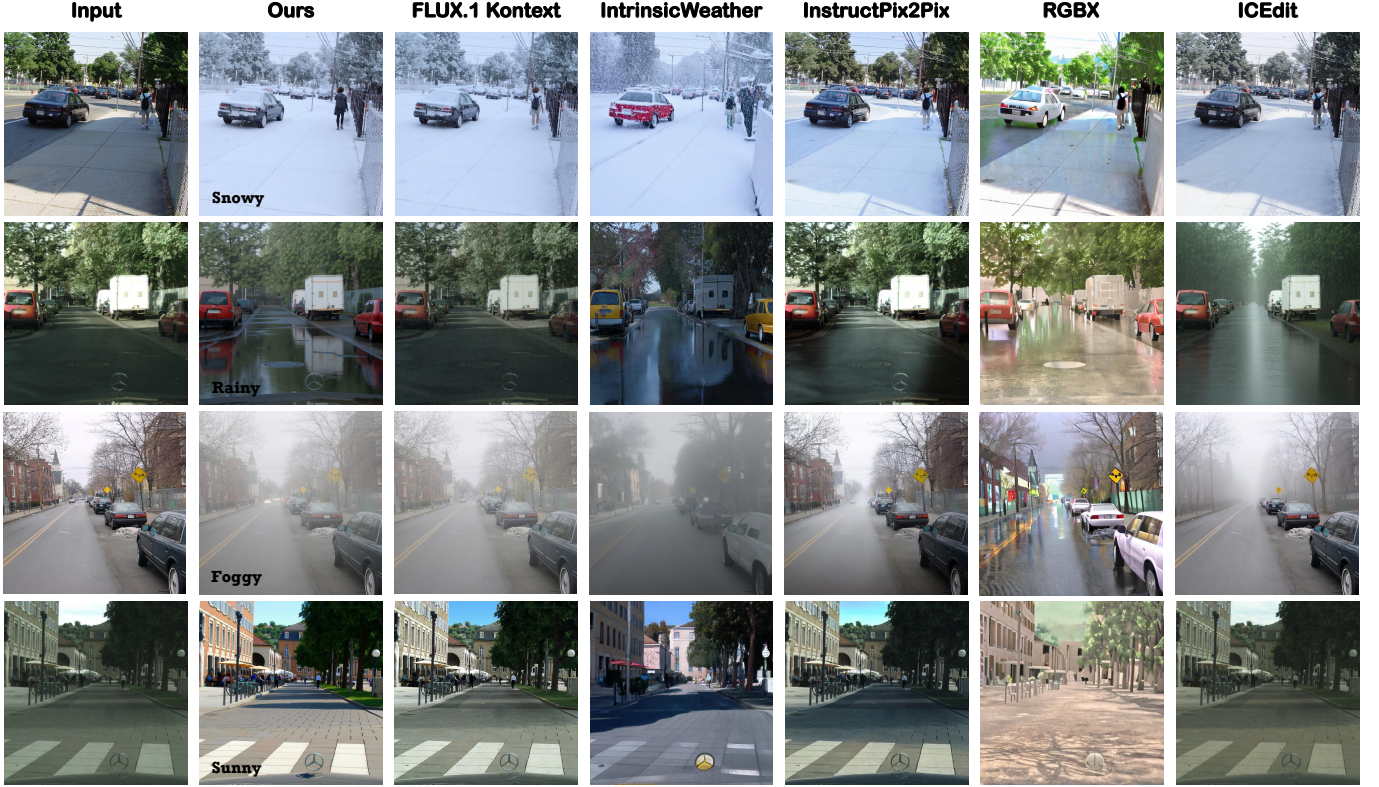


Fig. 3. **Comparison with different baselines for real images.** Target weather prompts (“snowy”, “rainy”, “foggy”, “sunny”) are denoted in the bottom-left of our results. Conventional editing baselines struggle to preserve scene geometry and intrinsic material colors, frequently resulting in severe stylistic color shifts, illogical retained shadows, or structural collapse under extreme conditions. In contrast, our method effectively disentangles foreground semantics from environmental transformations. This ensures the strict preservation of object identities while accurately modifying only weather-related components, yielding physically grounded global relighting and accurate specular reflections.

the performance across three dimensions on a 0–10 scale: (1) **Success**, measuring the semantic alignment with the target weather instructions; (2) **Overedit**, evaluating the preservation of the original scene integrity and object identities; and (3) **Realism**, assessing overall photographic realism, physical consistency (e.g., lighting, shadows, and atmospheric effects), and artifact suppression. Furthermore, we incorporate PickScore [36] as a complementary metric to evaluate human-aligned subjective visual quality. Unlike traditional pixel-based metrics, PickScore is trained on massive human-preference datasets, allowing it to better capture high-level aesthetic quality and subjective visual appeal.

B. Qualitative Results

As illustrated in Fig. 3, PC-Weather achieves a favorable balance between instruction fidelity and source-scene preservation across all four weather conditions. Rather than imposing a uniform appearance shift, our method applies region-dependent weather responses while retaining geometry-sensitive scene content.

In the snowy translation (first row), FLUX.1 Kontext [12] renders snowfall and accumulation but leaves the pedestrian largely unchanged, producing a semantic inconsistency between the wintry environment and the human appearance. IntrinsicWeather [13] changes the central vehicle from black to red, violating its intrinsic material identity. The remaining baselines either under-edit the scene or retain strong

cast shadows and illumination cues from the original sunny condition. In contrast, PC-Weather jointly adapts the road, vegetation, vehicles, and pedestrian to the snowy condition while preserving the original layout and vehicle identity.

In the rainy translation (second row), IntrinsicWeather and ICEdit [31] produce a pronounced rainy appearance, but substantially alter vehicle appearance and scene structure. Other baselines fail to fully remove clear-weather cues, most notably the pre-existing sunlight and hard shadows. Our result instead combines visible rainfall with a wet road surface and spatially coherent reflections, while maintaining the positions and geometry of vehicles, buildings, and the roadway.

The foggy example (third row) further highlights the physical awareness of our regional formulation. PC-Weather preserves the scene structure while producing depth-dependent

TABLE I
QUANTITATIVE COMPARISON OF WEATHER EDITING PERFORMANCE ON REAL-WORLD TRAFFIC SCENES

Method	Success ↑	Over edit ↑	Realism ↑	Pick Score ↑
RGB↔X [19]	4.455	3.441	2.786	19.985
FLUX.1 Kontext [12]	<u>9.214</u>	8.828	7.726	20.770
InstructPix2Pix [1]	5.737	7.757	6.348	20.025
ICEdit [31]	8.463	7.913	6.909	20.814
IntrinsicWeather [13]	8.654	6.196	6.078	21.193
Ours	9.640	<u>8.318</u>	<u>7.647</u>	<u>20.948</u>

visibility attenuation, with distant content losing contrast more strongly than nearby vehicles and road markings, rather than merely overlaying a spatially uniform gray veil. It also turns on the distant vehicle’s lights as a context-appropriate response to reduced visibility, a detail absent from the FLUX.1 Kontext result. Finally, in the sunny translation (fourth row), PC-Weather yields the most convincing illumination change among the compared methods, introducing natural brightness and coherent directional shadows while keeping the source geometry intact. Together, these examples demonstrate that PC-Weather edits weather-sensitive appearance and semantics where necessary without sacrificing the material identity and spatial structure of safety-critical traffic content.

C. Quantitative Results

Table I reports complementary measures of instruction fulfillment, content preservation, physical realism, and human-aligned visual preference on 492 traffic-scene images. Higher values indicate better performance for all four metrics. Best and second-best results are shown in bold and underline, respectively. PC-Weather is the only method that ranks among the top two on all four metrics, demonstrating a consistently strong balance across these competing objectives rather than optimizing a single aspect of editing quality.

In particular, PC-Weather achieves the best **Success** score of 9.640, exceeding the strongest baseline, FLUX.1 Kontext, by 0.426. This clear margin indicates that our region-specific formulation more reliably realizes the requested weather condition. Meanwhile, PC-Weather ranks second on both **Overedit** (8.318) and **Realism** (7.647), trailing FLUX.1 Kontext by only 0.510 and 0.079, respectively. The small realism gap, together with the substantially higher Success score, shows that the improvement in instruction fidelity is achieved without sacrificing photographic quality. Moreover, the strong Overedit result confirms that the method does not obtain its editing success through indiscriminate scene rewriting; instead, intrinsic anchoring and spatially restricted velocity corrections preserve object identity and traffic-scene structure while permitting the required weather changes.

PC-Weather further obtains the second-best **PickScore** [36] of 20.948, only 0.245 below IntrinsicWeather and higher than the remaining baselines. This human-preference-aligned result complements the MLLM evaluation and confirms that the favorable trade-off between effective editing and content preservation also translates into competitive perceptual quality. Notably, these results are achieved in a fully training-free manner, without LoRA fine-tuning or task-specific optimization, highlighting the practical generalization of PC-Weather across real-world traffic scenes.

D. User Study

To complement the automated evaluations, we conducted a forced-choice user study with 17 participants, who completed a total of 784 judgments in a randomized, side-by-side setup. In each trial, participants selected the overall preferred result by jointly considering visual realism, instruction fidelity, and artifact suppression. We report the **Preference Rate**, i.e., the

TABLE II
USER PREFERENCE RATES (%) BY WEATHER CONDITION

Method	Weather			Overall
	Snowy	Rainy	Sunny	
FLUX.1 Kontext [12]	26.2	11.0	22.3	19.8
InstructPix2Pix [1]	1.9	11.0	5.6	6.2
ICEdit [31]	35.4	23.4	1.6	20.1
Ours	36.5	54.6	70.5	53.9

TABLE III
QUANTITATIVE ABLATION STUDY OF INTRINSIC ANCHORING (IA) AND REGION-AWARE VELOCITY COMPOSITION (RVC)

Method	Success $\uparrow$	Over edit $\uparrow$	Realism $\uparrow$	MLLM	
				Avg. $\uparrow$	PickScore $\uparrow$
PC-Weather	9.610	7.860	6.920	8.130	21.518
w/o IA	9.160	8.080	6.980	8.073	21.505
w/o RVC	9.490	7.860	6.820	8.057	21.469

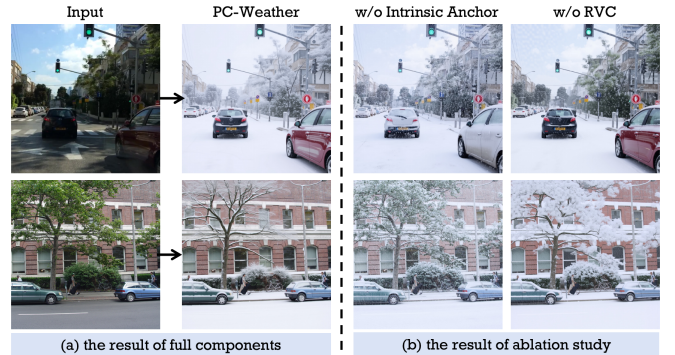


Fig. 4. **Qualitative ablation study of intrinsic anchoring and RVC.** Removing the intrinsic anchor causes material-identity drift, while removing RVC leads to under- or over-editing across semantic regions. The full model achieves physically plausible weather transformation while preserving object identity and scene structure.

percentage of judgments in which each method was selected, for snowy, rainy, and sunny translations; higher rates indicate stronger preference. The study protocol was completed before IntrinsicWeather and the foggy condition were added to our expanded experiments; consequently, neither is included in this user study. The **Overall** score is the arithmetic mean of the three weather-specific preference rates.

As shown in Table II, PC-Weather achieves the highest preference rate under every evaluated weather condition. Its overall preference reaches 53.9%, exceeding the second-best ICEdit by 33.8 percentage points. The snowy case is more competitive, where PC-Weather leads ICEdit by 1.1 points (36.5% vs. 35.4%). The advantage becomes pronounced for rainy and sunny translations: PC-Weather outperforms the respective runner-up by 31.2 and 48.2 points. These results indicate that the improvements in instruction adherence and structural preservation observed in the automated evaluation are also consistently perceived by human observers.

E. Ablation Study

We conduct complementary quantitative and qualitative ablations to examine the roles of intrinsic anchoring (IA) and



Fig. 5. **PC-Weather-synthesized training images for downstream detection.** Representative Cityscapes images translated to snow, rain, fog, and sandstorm (1,500 per domain; 6,000 in total). Red boxes show vehicle predictions from a detector trained on these synthetic images.

Region-Aware Velocity Composition (RVC).

Quantitative ablation. Table III reports results on 100 images randomly sampled from the three evaluation datasets; MLLM Avg. is the arithmetic mean of Success, Overedit, and Realism. The full PC-Weather model achieves the best Success (9.610), MLLM Avg. (8.130), and PickScore (21.518). Removing IA increases the individual Overedit and Realism scores (8.080 vs. 7.860 and 6.980 vs. 6.920), but reduces Success by 0.450, MLLM Avg. by 0.057, and PickScore by 0.013. This pattern highlights the trade-off between conservative preservation and complete target-weather fulfillment. Removing RVC retains the Overedit score but decreases Success, Realism, MLLM Avg., and PickScore, indicating that region-aware composition improves weather fidelity and perceptual quality without additional content drift.

Qualitative ablation. Figure 4 provides complementary visual evidence. Removing the intrinsic anchor weakens the preservation of weather-invariant material and semantic attributes. In the top row, the central black vehicle is incorrectly changed to white; in the bottom row, the tree retains an implausibly dense foliage structure that is merely whitened, rather than undergoing a physically plausible winter transition. These examples indicate that the intrinsic reference supports material identity and structural boundaries.

Removing RVC produces inconsistent regional editing strengths. The top-row result under-edits weather-sensitive regions, leaving visible green vegetation, whereas the bottom-row result over-edits the tree with excessive snow masses that obscure its branch structure. In contrast, the full model preserves vehicle colors and scene geometry while applying appropriate weather responses to vegetation and exposed surfaces. Together, the quantitative and qualitative evidence supports the complementary roles of intrinsic anchoring in stabilizing material and structural identity and RVC in adapting where and how strongly weather-related changes are applied.

F. Downstream Tasks on Real Adverse-Weather Data

Image-level scores do not establish whether edited images are useful as training data. We therefore assess *task utility*: a vehicle detector is trained only on clear Cityscapes images or their synthetic adverse-weather counterparts, and is then

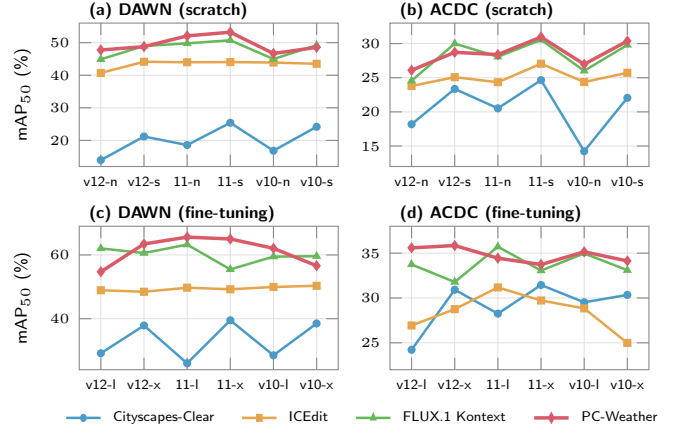


Fig. 6. **Model-wise mAP₅₀ on real adverse-weather test sets.** Each panel compares identical detector variants trained with clear Cityscapes or one of three synthetic datasets. PC-Weather is best in 18 of 24 configurations; the remaining cases make the architecture-dependent variation visible rather than hiding it behind the averages in Table IV.

evaluated, without target-domain adaptation, on real adverse-weather images from DAWN and ACDC. Better cross-dataset detection indicates that the generated images preserve label-relevant geometry while reducing the appearance gap to real adverse weather.

1) *Protocol: Training data and supervision.* We randomly sample 1,500 clear-weather images from Cityscapes [30]. PC-Weather, FLUX.1 Kontext, and ICEdit each translate every source image to rain, snow, fog, and sandstorm, yielding 6,000 images per method (Fig. 5); the unchanged 1,500 images form the *Cityscapes-Clear* baseline. A fixed pretrained YOLO detector produces vehicle pseudo-labels on the clear source images, and the same labels are transferred to all three corresponding synthetic sets. Thus, source identities and supervision are controlled across methods; both weather realism and any synthesis-induced geometric drift are reflected in downstream performance.

Training and evaluation. We use two complementary protocols across three YOLO generations. In the *scratch* protocol, compact YOLOv12-/YOLO11-/YOLOv10-*n/s* models are trained without pretrained weights, directly testing the supervision supplied by each dataset. In the *fine-tuning* protocol, pretrained *l/x* variants are adapted using the same data, testing whether synthetic weather remains useful with a stronger initialization. Every model is evaluated on the real DAWN [37] and ACDC [38] test sets; no target-set image or annotation is used for training. We report mAP_{50:95}, mAP₅₀, Precision, and Recall using a common configuration (confidence 0.001, IoU 0.5, 640-pixel input, and at most 300 detections). Table IV averages each metric over the six detector variants in a protocol; the best result for each protocol and test set is shown in bold.

2) *Cross-Dataset Results: Scratch training.* PC-Weather gives the highest mean AP on both test sets. On DAWN, it reaches 0.3083 mAP_{50:95} and 0.4950 mAP₅₀, absolute gains of 0.0070 and 0.0141 over FLUX.1 Kontext. The margin is smaller on ACDC (0.0003 and 0.0042), where FLUX.1 Kontext also has 0.0034 higher Precision; nevertheless, PC-

TABLE IV
CROSS-DATASET VEHICLE DETECTION ON REAL ADVERSE-WEATHER TEST SETS

Protocol	Test Set	Training Data	mAP _{50:95}	mAP ₅₀	Precision	Recall
Scratch	DAWN	Cityscapes-Clear	0.1138	0.2004	0.4648	0.1745
		ICEdit	0.2623	0.4338	0.6924	0.3793
		FLUX.1 Kontext	0.3013	0.4809	0.7188	0.4185
		PC-Weather	0.3083	0.4950	0.7196	0.4273
	ACDC	Cityscapes-Clear	0.1113	0.2050	0.3789	0.3030
		ICEdit	0.1361	0.2506	0.4221	0.3528
		FLUX.1 Kontext	0.1667	0.2816	0.4441	0.3793
		PC-Weather	0.1670	0.2858	0.4407	0.3888
Fine-tuning	DAWN	Cityscapes-Clear	0.2072	0.3326	0.6294	0.2639
		ICEdit	0.3048	0.4943	0.7343	0.4425
		FLUX.1 Kontext	0.3778	0.6007	0.7562	0.5370
		PC-Weather	0.3826	0.6124	0.7657	0.5430
	ACDC	Cityscapes-Clear	0.1728	0.2912	0.4430	0.4115
		ICEdit	0.1627	0.2840	0.4491	0.3929
		FLUX.1 Kontext	0.2105	0.3373	0.4794	0.4582
		PC-Weather	0.2190	0.3481	0.4877	0.4672

Weather retains the best mAP_{50:95}, mAP₅₀, and Recall. Relative to ICDIT, PC-Weather improves mAP_{50:95} by 17.54% on DAWN and 22.70% on ACDC. These results show that the benefit is not tied to one adverse-weather benchmark.

Fine-tuning. Synthetic weather remains beneficial when adapting large pretrained detectors. PC-Weather achieves the best mean result for every metric on both datasets: 0.3826/0.6124 mAP_{50:95}/mAP₅₀ on DAWN and 0.2190/0.3481 on ACDC. Against FLUX.1 Kontext, the relative mAP_{50:95} gains are 1.27% and 4.04%, respectively. The improvement over Cityscapes-Clear is substantially larger, confirming that model capacity and pretraining alone do not remove the adverse-weather domain gap.

Figure 6 exposes variation hidden by the averages. PC-Weather ranks first in mAP₅₀ for 18 of the 24 detector–dataset configurations, including at least four of six large models on each test set. The advantage is therefore broad but not uniform: FLUX.1 Kontext is stronger for several individual variants, most notably YOLOv12-*l* on DAWN. Reporting averages over three detector generations avoids selecting a favorable architecture and supports the more measured conclusion that PC-Weather is the most consistent training source overall.

Across all 24 configurations, PC-Weather averages 0.2692 mAP_{50:95} and 0.4354 mAP₅₀. These correspond to relative gains of 1.96% and 2.40% over FLUX.1 Kontext, 24.37% and 19.06% over ICDIT, and 77.98% and 69.22% over Cityscapes-Clear. Because source images and pseudo-labels are identical, these gaps capture the joint effect of target-weather realism and label-preserving structure. The downstream evidence thus complements the perceptual evaluation: PC-Weather produces not only plausible edits, but synthetic data that transfers more effectively to real adverse-weather detection.

G. Limitations

PC-Weather has two main limitations. First, intrinsic anchoring and regional instructions encourage physically plausible edits but do not explicitly enforce scene geometry or atmospheric transport, limiting guarantees for depth-dependent

effects such as dense fog. Second, our evaluation is restricted to single-image traffic scenes and vehicle detection, leaving temporal consistency and broader domain and task generalization unverified. Incorporating geometric and temporal priors could address these limitations.

V. CONCLUSION

We presented PC-Weather, a training-free framework for physically motivated and semantically controllable weather editing in traffic scenes. Our core contributions are intrinsic anchoring, which uses estimated albedo as an illumination-reduced preservation reference, and Region-Aware Velocity Composition, which adaptively fuses category-specific residual trajectories around a global reference field. By refining a global weather request into category-specific instructions, PC-Weather replaces a single undifferentiated editing request with semantically grounded regional responses. Extensive experiments demonstrate that this formulation yields high-fidelity weather edits, preserves label-relevant scene structure, and improves the utility of synthesized data for adverse-weather perception.

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